Vrushang Anand

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Education

University of California, Irvine *Jun 2026*
Bachelor of Science, Computer Engineering *GPA: 3.5*

Experience

Resilient Cyber-Physical Systems Lab - Robotics Lab
AI Researcher *Jun 2026 - Present*

- Redesigned the state input and attention architecture of a vision-language-action model, raising task success 2.4× on bimanual manipulation.
- Built a JAX pipeline to train and evaluate 20 multi-seed runs across 12,000 simulated episodes on a single GPU.
- Augmented a robot demonstration dataset with object positions recovered through MuJoCo physics replay, to about 2 mm.

Undergraduate Researcher *Mar 2025 - Jun 2026*

- Developed safety-constrained controllers for autonomous systems using backward reachability analysis.
- Engineered a CUDA-accelerated PyTorch pipeline for Bernstein polynomial coefficients, reducing time complexity from $O(n^{2l})$ to $O(n^{l+1})$.
- Integrated a custom neural network with Bernstein polynomial activation layers into Stable-Baselines3 (PPO, A2C).

HyperXite - SpaceX Hyperloop Competition
Control Systems Lead *Jun 2024 - Jun 2025*
Control Systems Engineer *Aug 2023 - Jun 2024*

- Led the control-systems sub-team to Best Guidance System, Best Research, and Best Presentation awards at Hyperloop Global Conference 2025.
- Architected a real-time control system in C++ and Rust coordinating propulsion, pneumatic, braking, and thermal subsystems across distributed embedded hardware.
- Built an end-to-end telemetry and remote controlled system over WebSockets, streaming diagnostics to a ground control station for live monitoring and pod control.

SwipeTax - Fintech Startup *Sep 2024 - Mar 2025*
Software Engineering Intern

- Designed and implemented a swipe-based expense classification system, reducing manual bookkeeping time from several hours to a few minutes and simplifying separation of personal vs. business expenses for small business users.
- Built and deployed backend services in Node.js on AWS (EC2 and S3) for transaction ingestion and processing across distributed components with Plaid API for real-time financial data synchronization.
- Shipped a scalable full-stack accounting platform automating core financial workflows.

Projects

TISA: Tensor Instruction Set Architecture | Technical Report
Python, C++, Verilog, ONNX

- Designed and implemented a custom architecture for deploying trained neural networks onto FPGAs, reducing the need to redesign hardware for each model and simplifying hardware accelerated AI deployment.
- Built a compiler that translates neural-network models into instructions for the FPGA, handling model analysis, memory organization, and hardware-specific execution.
- Designed the architecture with separate units for loading data, performing calculations, and storing results, allowing data movement and computation to run in parallel.

Reinforcement-Learning Linux Kernel Scheduler | GitHub | Article
C, Python, Linux Kernel, PyTorch

- Designed and implemented a reinforcement-learning-based Linux CPU scheduler that improved average turnaround time in 70% of the benchmark workloads compared to the standard Linux scheduler.
- Built a simulator that recreated Linux CPU scheduling across different workloads and used it to train and evaluate the neural network before deploying the model into the kernel.
- Integrated the trained model into the Linux kernel with a 15 µs fixed-point C inference path.

Skills

Programming: C, C++ (11/14/17/20+), Python, Rust, JavaScript/TypeScript

Machine Learning: PyTorch, NumPy, JAX, Stable-Baselines3, OpenAI Gym

Hardware: Verilog, FPGA, SPI/I2C/GPIO/CAN, SBC, ESP32

Web Development: ReactJS, Next.js, Tailwind CSS

Cloud: AWS, GCP, Supabase, Vercel

Tools: Git, Docker, QEMU, GDB, Bash, Linux/UNIX